

FAIRHIRE AI: MITIGATING BIAS IN HIRING MODELS

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INTRODUCTION

Background

- AI/ML are widely used in recruitment for resume screening and candidate selection
- While it improves efficiency it can inherit and amplify bias from historical data
- Risk of discrimination based on gender, ethnicity, or socioeconomic background
- Raises ethical, legal, and societal concerns
- Even high-performing models can produce unfair outcomes

Research Question

- To what extent do machine learning models exhibit measurable bias in hiring decisions and how effectively can mitigation strategies reduce this bias without significantly impacting performance?



LITERATURE REVIEW

Related Work & Existing Approaches

Prior Research

- Studies show ML systems can replicate existing societal biases
- Fairness often conflicts with predictive accuracy

Key Works

- *Fairness in Machine Learning*
 - Discusses fairness challenges in AI systems
- *Inherent Trade-Offs in the Fair Determination of Risk Scores*
 - Explores trade-offs between fairness and accuracy

Existing Tools

- IBM AIF360 Toolkit
 - Measures and mitigates bias in ML models

Our Contribution

- Compare multiple fairness metrics
- Analyze intersectional bias (gender + ethnicity)
- Evaluate:
 - Pre-processing mitigation
 - In-processing mitigation
 - Post-processing mitigation
- Study fairness vs. performance trade-offs



OBJECTIVES

Primary Goal

- Develop and evaluate fairer AI-driven hiring models

Objectives

- Build baseline ML hiring models:
 - Logistic Regression
 - Decision Tree
 - Random Forest + SMOTE
 - Gradient Boosting
- Measure bias using fairness metrics:
 - Demographic Parity Difference (DPD)
 - Equal Opportunity Difference (EOD)
- Apply bias mitigation techniques:
 - Reweighting
 - Exponentiated Gradient Fairness Constraint
 - Threshold adjustment
- Compare:
 - Model accuracy
 - Fairness improvements
 - Performance trade-offs

Expected Outcome

- Reduce measurable hiring bias
- Maintain acceptable predictive performance
- Provide insights into fairness-aware AI systems



DATASET

Dataset Used: *Hiring Discrimination in the Organisational Context*

- EU Horizon Europe dataset
- 15,036 observations
- 103 features

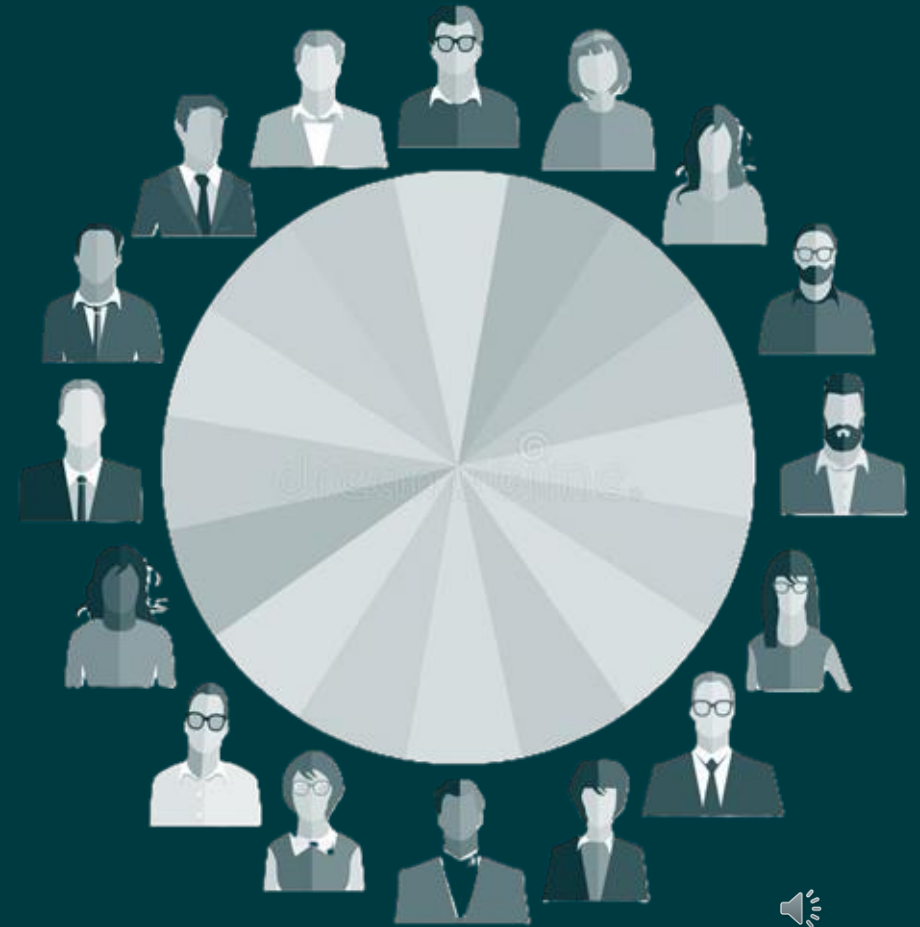
Protected Attributes

- Gender
- Ethnicity/Nationality
- Parenthood Status
- Relationship Status
- Language Proficiency
- Education

Target Variable

- Hire = 1
- No Hire = 0

Class	Count
No Hire	12,530
Hire	2,506



METHODOLOGY

1. Data Preprocessing

- Created a binary hiring outcome (Hire vs. No Hire)
- Encoded categorical variables using one-hot encoding
- Split data into 80% training / 20% testing
- Addressed class imbalanced using class weighting, reweighing, SMOTE oversampling
- After SMOTE 10,001 samples per class

2. Models

Logistic Regression

- Classification Model
- Used as baseline model
- Highly interpretable and commonly used in hiring analytics

Decision Tree

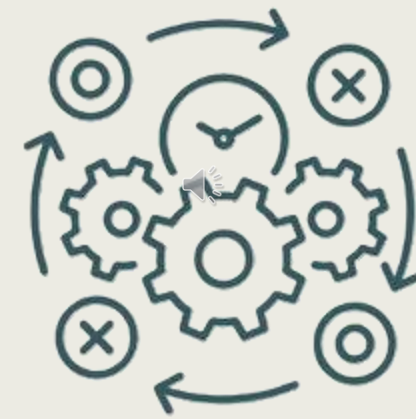
- Rule-based classification model
- Captures nonlinear relationships
- Provides explainable decision paths

Random Forest + SMOTE

- Ensemble of decision trees
- Uses SMOTE to balance minority hiring outcomes
- Improves robustness to class imbalance

Gradient Boosting

- Sequential ensemble model
- Corrects errors made by previous trees
- Often achieves strong performance on tabular datasets



METHODOLOGY

3. Fairness Evaluation

Demographic Parity Difference (DPD)

- Measures differences in hiring prediction rates across demographic groups

Equal Opportunity Difference (EOD)

- Measures differences in true positive rates across demographic groups
- Lower values indicate fairer outcomes

4. Bias Mitigation Methods

Pre-Processing

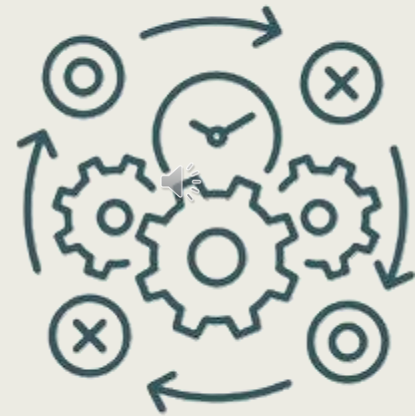
- Reweighting: Adjusts training sample weights to reduce demographic disparities before model training

In-Processing

- Exponentiated Gradient Fairness Constraint: Incorporates fairness into model optimization

Post-Processing

- Threshold Adjustment: Modifies decision thresholds after training to improve fairness outcomes



METHODOLOGY

5. Performance Comparison

Evaluate Models using:

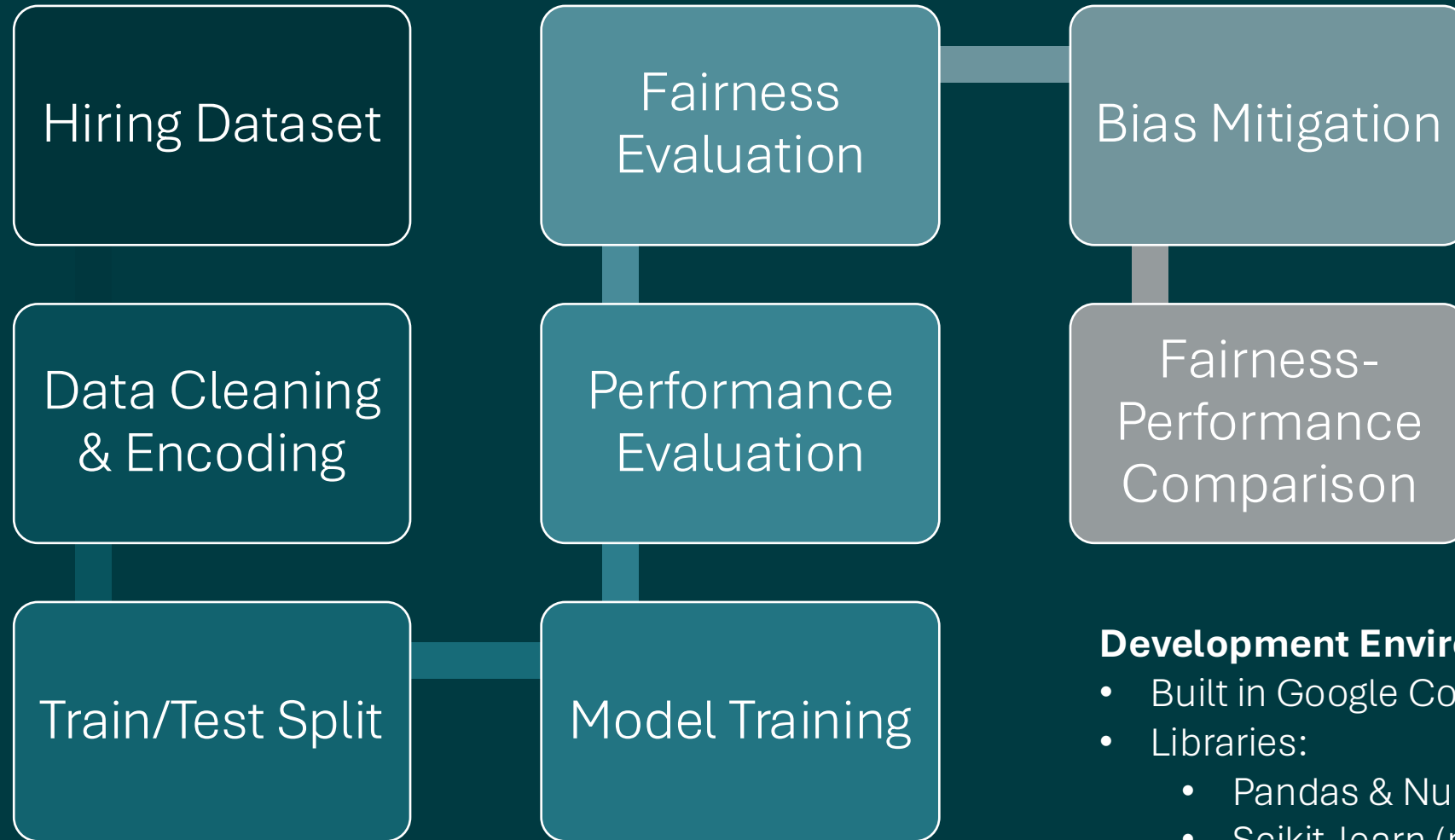
- Accuracy
- Precision
- Recall
- F1-score

6. Tools & Libraries

- Python
- Scikit-learn
- Fairlearn / AIF360
- Pandas & Numpy
- Matplotlib



IMPLEMENTATION



Development Environment

- Built in Google Colab using Python
- Libraries:
 - Pandas & NumPy (data processing)
 - Scikit-learn (machine learning models)
 - Fairlearn (fairness metrics and mitigation)
 - Matplotlib (visualizations)



BASELINE MODEL RESULTS

Model	Accuracy	F1
Logistic Regression	0.686	0.724
Decision Tree	0.651	0.697
Random Forest +SMOTE	0.779	0.773
Gradient Boosting	0.779	0.773

- Gradient Boosting achieved the highest accuracy and F1 score
- Decision Tree produced the weakest performance
- Random Forest + SMOTE improved performance over Logistic Regression
- Logistic Regression remained the fairness baseline due to interpretability



FAIRNESS ANALYSIS

Demographic Parity Difference

Attribute	DPD
Gender	0.032
Ethnicity	1.000
Parenthood	0.003
Relationship	0.013
Language Level	0.513
Education	0.519

Equal Opportunity Difference

Attribute	EOD
Gender	0.038
Ethnicity	1.000
Parenthood	0.002
Relationship Status	0.030
Language Level	0.719
Education	0.751

Gender x Ethnicity

Group	Hire Rate
Man + Native	1.000
Woman + Native	1.000
Woman + Ukrainian	0.071
Man + Ukrainian	0.000
Man + Other Ethnic Group	0.000
Woman + Other Ethnic Group	0.000

Demographic Parity Difference

- Largest disparities in DPD were found in ethnicity and language proficiency
- Smallest disparities in DPD were found in parenthood, relationship status, and gender

Equal Opportunity Difference

- Language proficiency and ethnicity had the largest impact on hiring predictions according to EOD

Gender x Ethnicity

- Maximum disparity gap = 1.0
- Bias becomes more severe when demographic attributes are combined



BIAS MITIGATION RESULTS

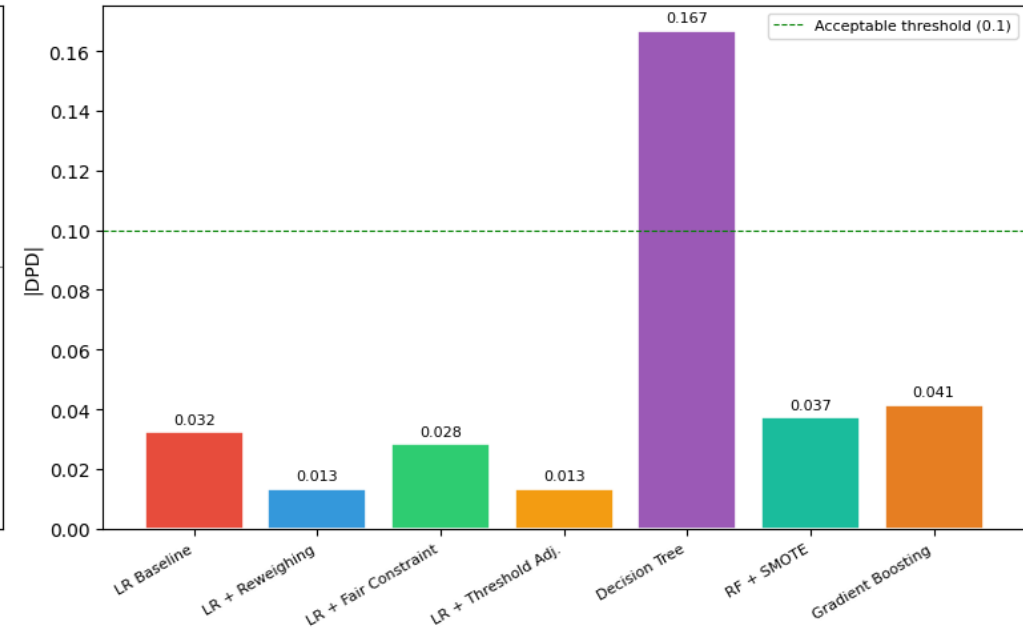
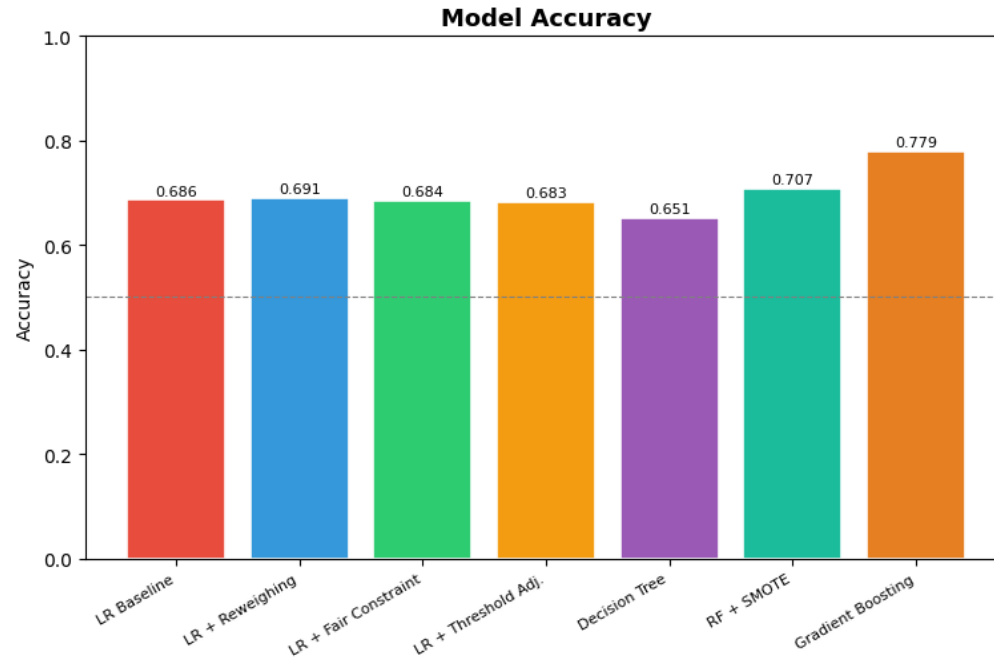
Model	Accuracy	DPD	EOD
Baseline LR	0.686	0.032	0.038
Reweighting	0.691	0.013	0.054
Fair Constraint	0.686	0.029	0.038
Threshold Adj.	0.692	0.026	0.045

- Reweighting achieved lowest DPD
- Threshold Adjustment achieved highest accuracy among mitigation methods
- Fairness improved with minimal performance loss

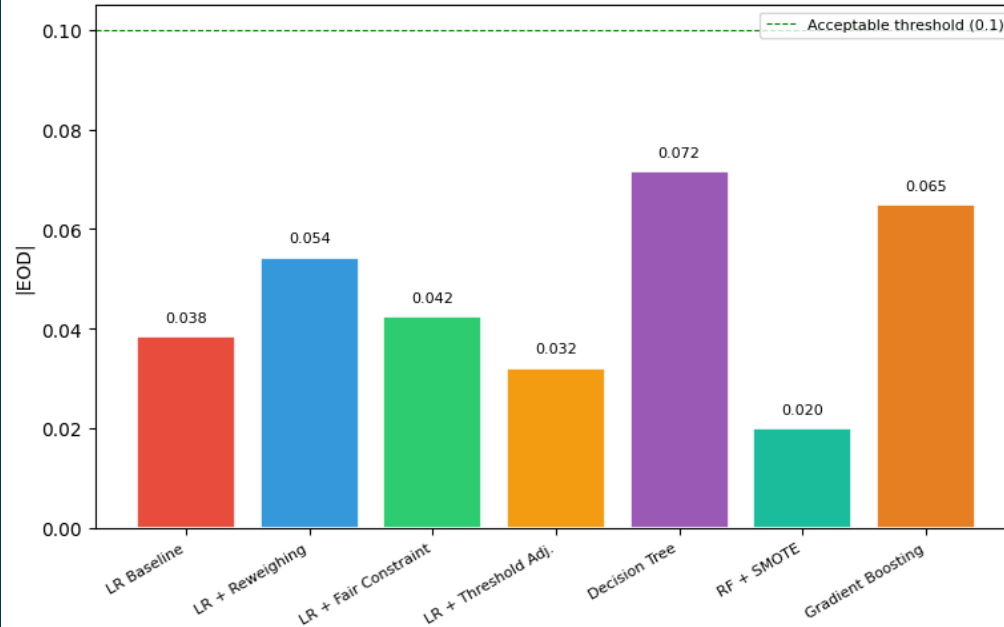


FairHire AI — Fairness vs Performance Results

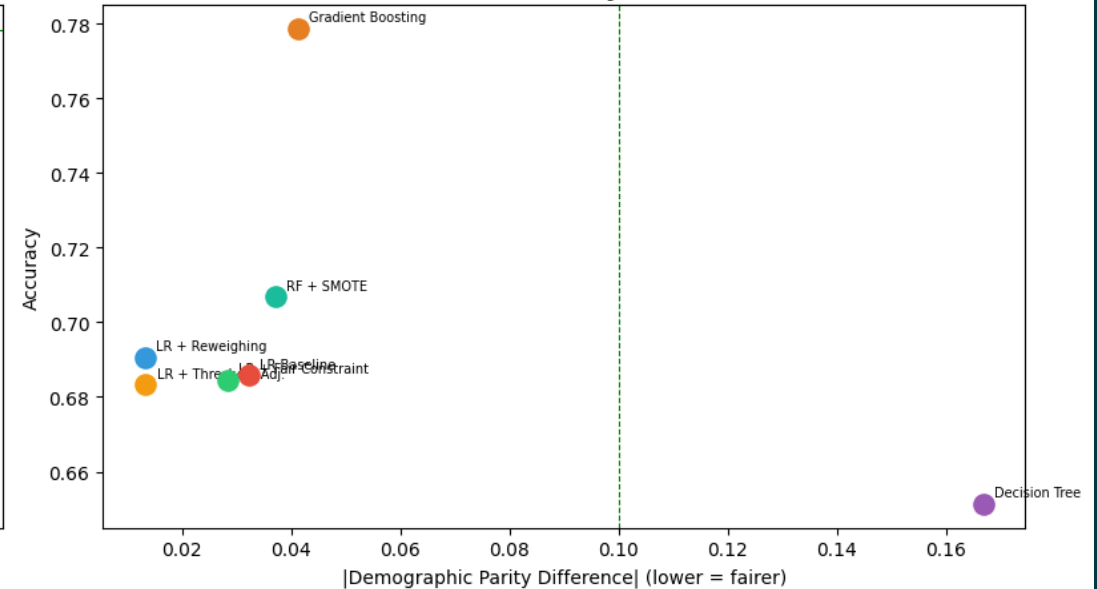
[Demographic Parity Difference]
(lower = fairer)



[Equal Opportunity Difference]
(lower = fairer)



Fairness-Accuracy Trade-off



CHALLENGES & LIMITATIONS

CHALLENGES

Imbalanced dataset

- Only about 17% of observations were positive hiring outcomes
- Baseline models tended to favor the majority class making accuracy alone misleading

Fairness Measurement

- Different fairness metrics can produce different conclusions
- A model may appear fair according to one metric while showing disparities under another

Multiple Protected Attributes

- Evaluating fairness across gender, ethnicity, language proficiency, parenthood, relationship status increased the complexity of analysis

LIMITATIONS

Survey-Based Data

- The dataset contains hiring recommendations from a factorial survey experiment rather than actual hiring decisions
- Results may not fully reflect real-world recruitment behavior

Limited Geographic Scope

- Data was collected from European countries
- Findings may not generalize to labor markets in the United States or other regions

Restricted Demographic Coverage

- The study focuses on attributes available in the dataset
- Other potentially important factors weren't examined

Model Scope

- Only Logistic Regression, Decision Tree, and Random Forest + SMOTE, and Gradient Boosting were evaluated
- More advanced models may exhibit different fairness-performance trade-offs such as deep neural networks, transformers, and large language models





FUTURE WORK

Expand to Real-World Hiring Data

- Evaluate fairness using actual recruitment and hiring records rather than survey-based hiring recommendations.
- Compare whether biases observed in experimental settings appear in real hiring decisions.

Explore Additional Fairness Metrics

- Incorporate metrics such as:
 - Equalized Odds
 - Predictive Parity
 - Calibration
- Provide a more comprehensive assessment of fairness.

Evaluate More Advanced Models

- Investigate fairness in:
 - Deep Neural Networks
 - Large Language Models used in resume screening
- Compare their fairness-performance trade-offs with traditional ML models.

Expand Intersectional Analysis

- Examine additional combinations of demographic attributes.
- Better understand how multiple characteristics interact to influence hiring outcomes.

Broader Geographic and Demographic Coverage

- Apply the methodology to datasets from the United States and other regions.
- Study additional protected characteristics and labor market contexts.



CONCLUSION

Key Findings

- Machine learning hiring models exhibited measurable bias across several demographic attributes
- The largest disparities were observed for ethnicity and language proficiency, while gender, parenthood, and relationship status showed smaller differences
- Intersectional analysis revealed that combining demographic characteristics can expose larger disparities than examining a single attribute alone
- Fairness mitigation techniques successfully reduced bias while maintaining similar predictive performance

Best Performing Approaches

- Gradient Boosting achieved the highest predictive performance (77.9%)
- Reweighing achieved the strongest DPD (0.0132)
- Random Forest + SMOTE achieved the lowest EOD (0.0200)

Overall Takeaway

- Machine learning hiring systems can exhibit measurable bias
- Fairness mitigation techniques substantially reduce disparities
- The best model depends on whether fairness or predictive performance is prioritized



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Q&A

